For Examiner's Use

7 Many devices are designed to create a spray of tiny droplets. The effectiveness of these devices usually depends on droplet size. One example is an agriculture pesticide spray in which a few large droplets do not coat the leaves of plants as well as many small droplets. Another example is a fuel injection system for an engine.

Measuring the size of droplets present in a spray is difficult to do by direct means but instruments called droplet sizers can be purchased which make droplet sizing a fast routine operation.

The principle of operation of one such sizer is shown in Fig. 7.1, in which light from a helium/neon laser is passed through a spray of droplets of uniform diameter and forms a circular diffraction ring of radius *x*. The diameter *d* of the droplets is related to *x* by the equation

$$d = k\frac{\lambda}{x}$$

In this equation λ, the wavelength of the light, is 6.33×10^{-7} m. *k* is a constant equal to 0.474 m, and *d* and *x* are both in metres.

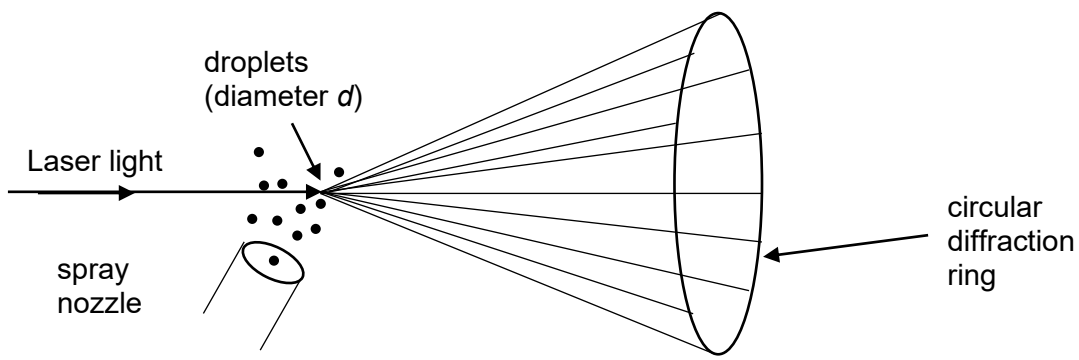

Fig. 7.1

In practice, a spray will consist of droplets of different sizes, so many rings of diffracted light will be caused. The diffraction pattern in Fig. 7.2a is projected on a flat surface containing many light sensitive detectors. The output from the detectors can be analysed by a computer and be shown in the form of a graph in Fig. 7.2b.

Fig. 7.2a

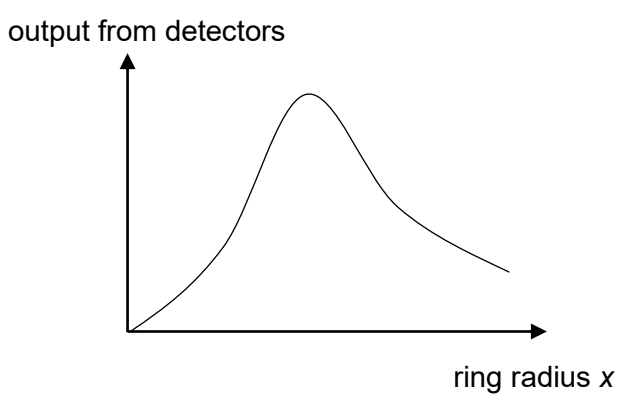

Fig. 7.2b

Examiner's Use

(a) Suggest two devices, other than those mentioned in the first paragraph of the passage, where droplet size is important.

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……[2]

(b) Outline a direct method for measuring droplet diameter.
Apparatus available includes: high speed camera, stroboscope, stopwatch, rulers.

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……[3]

(c) Give two reasons why direct methods are likely to be difficult for measuring droplets of small diameter.

……..

……[2]

(d) Calculate the value of *x* for a droplet of diameter

(i) 10 μm

x = …….. m [1]

(ii) 200 μm

x = …….. m [1]

(e) State whether a small value of x corresponds to large or to small droplets.

..... [1]

(f) Suggest, with reference to the equation given, how the radius of the circular diffraction ring can be increased for the same diameter of water droplets.

..... [1]

(g) Sketch on the axes in Fig. 7.3 curves to show the general shape of graphs that would be obtained if

(i) droplets with a wide range of diameters were used. Label as **(i)**. [1]

(ii) very small droplets with a narrow range of diameters were used. Label as **(ii)** [1]

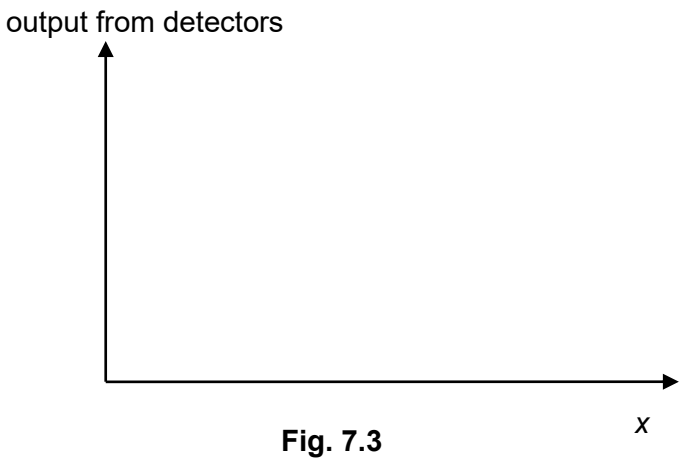

Fig. 7.3

Examiner's Use

(h) In practice, a cloud of spray droplets moves through the laser beam as shown at intervals in Fig. 7.4. The output from the detectors varies with time in the way shown in Fig. 7.5.

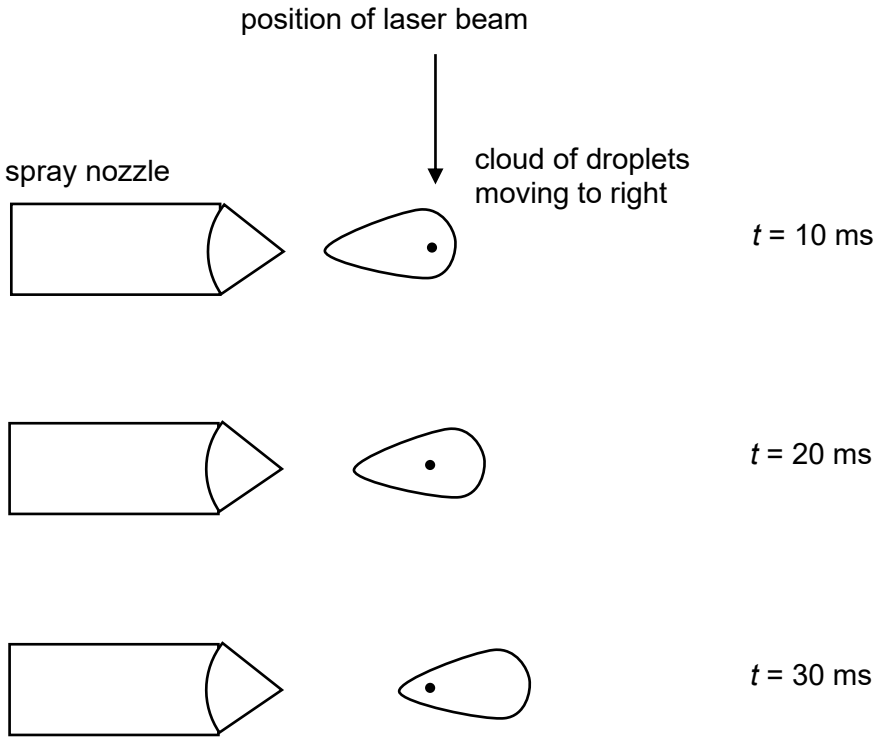

Fig. 7.4

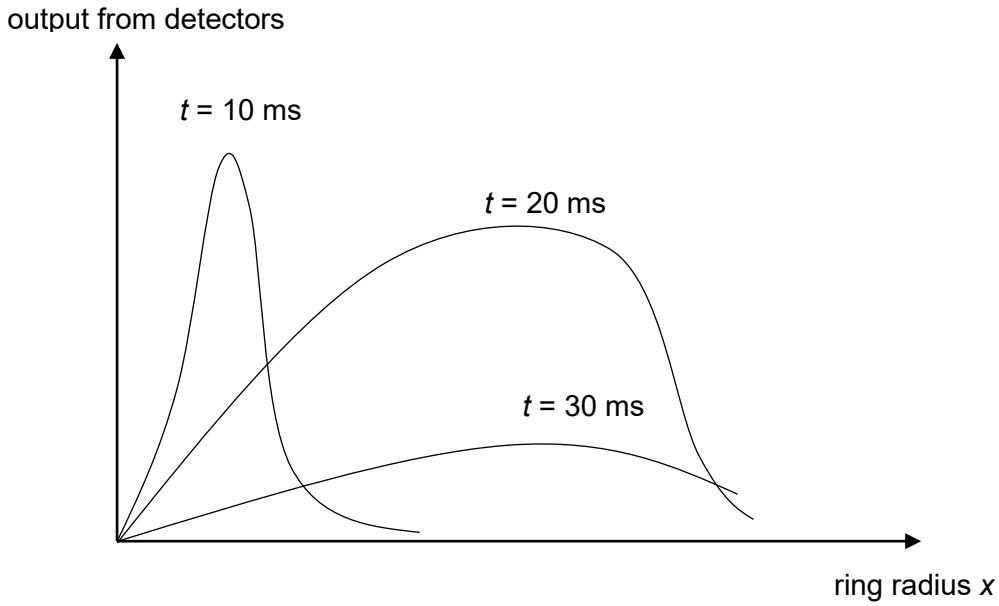

Fig. 7.5

With reference to Fig. 7.6, describe the distribution of droplets in the cloud according to their size and concentration (i.e. amount of droplets).

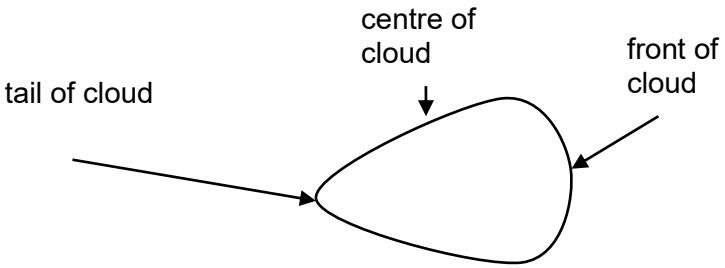

Fig. 7.6

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.....[6]

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